

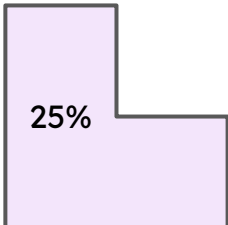


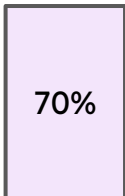
Problem solving with percentages and proportionality

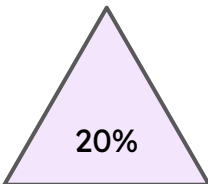
Task A: Using percentages to solve problems

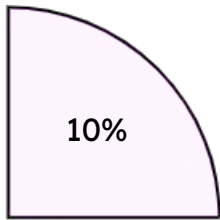
1) The images below show some parts of different shapes.

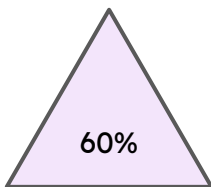
What might each whole shape look like?

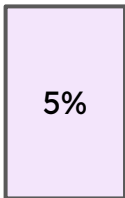
a) 

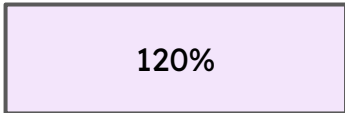
b) 

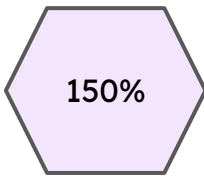
c) 

d) 

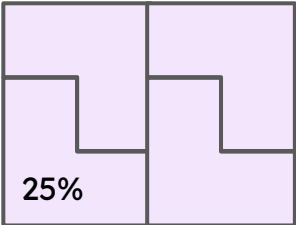
e) 

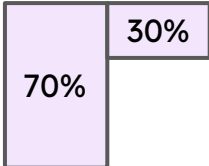
f) 

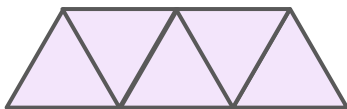
g) 

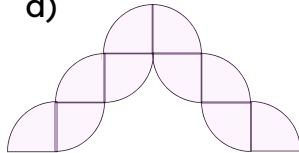
h) 

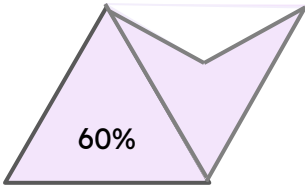
There are many possible answers. Below are some examples.

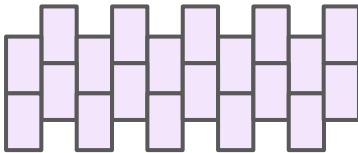
a) 

b) 

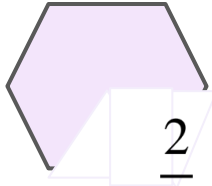
c) 
Each triangle is 20%

d) 
Each quarter circle is 10%

e) 

f) 

g) 
 $\frac{5}{6}$ of the original shape

h) 
 $\frac{2}{3}$ of the original shape

2) Aisha watched a TV program in which they buy items, mend or fix them and then sell the items.

She is going to rank the items based on their percentage profit or loss.

Item bought	Cost to buy (£)	Other costs (£)	Selling price (£)	profit/loss	% change
Vase	42	3	42.75	<i>loss</i>	<i>5</i>
Watch	75	10	93.50	<i>profit</i>	<i>10</i>
Toy	12	6	20.70	<i>profit</i>	<i>15</i>
Painting	38	17	50.60	<i>loss</i>	<i>8</i>
Chair	20	50	59.50	<i>loss</i>	<i>15</i>
Necklace	8	0	9.92	<i>profit</i>	<i>24</i>

Order: *chair, painting, vase, watch, toy, necklace*

Task B: Using proportionality to solve problems

1) A bag of sweets contains red, yellow, pink and green sweets.

There are 56 sweets in the bag.

25% of them are red.

$\frac{2}{7}$ of them are yellow.

The rest are pink and green in the ratio 5 : 8

How many of each colour are there?

$$\text{Red} = 14$$

$$\text{Yellow} = 16$$

$$\text{Pink} = 10$$

$$\text{Green} = 16$$



2) Aisha, Jun, Izzy, Sam and Laura raise money by doing a sponsored run.

They raise £600, Aisha raised 15%, Jun raised $\frac{4}{15}$, and Izzy raised $\frac{3}{8}$ of the total.

Sam and Laura collected the rest in the ratio 2 : 3

How much money did each of them raise?

$$\text{Aisha} = £90$$

$$\text{Jun} = £160$$

$$\text{Izzy} = £225$$

$$\text{Sam} = £50$$

$$\text{Laura} = £75$$

3) Andeep, Sofia and Jacob record how long it takes them to complete a puzzle.

Sofia's time is two-thirds of Andeep's time.

Jacob's time is 3 times Sofia's time.

The total of Jacob & Sofia's times is 88 seconds.

What is the difference between Andeep and Jacob's times?

$$\text{Andeep} = 33 \text{ seconds}$$

$$\text{Sofia} = 22 \text{ seconds}$$

$$\text{Jacob} = 66 \text{ seconds}$$

The difference between Andeep and Jacob's times is 33 seconds.

